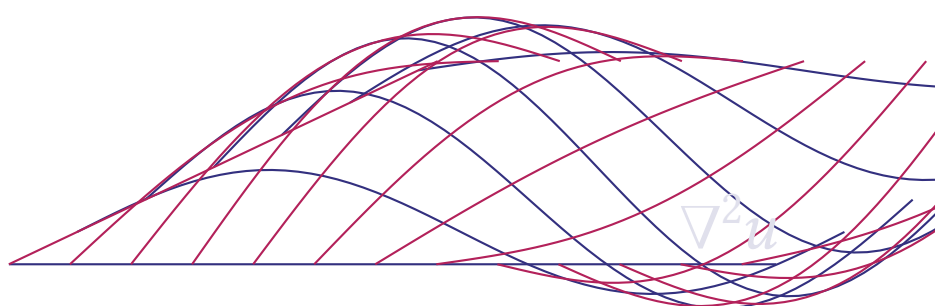


Two-Dimensional Wave and Heat Equations: Rectangle and Disc

معادلات دوبعدی موج و گرما: مستطیل و دایره



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13.1 The Two-Dimensional Wave Equation

Modelling a vibrating membrane (deriving the equation governing $u(x, y, t)$).

Physical assumptions:

- The mass is distributed uniformly over the membrane (a homogeneous membrane), and the membrane is perfectly flexible.
- The membrane is stretched so tightly that the force of gravity on it is negligible.
- The deflection of the membrane from equilibrium is very small.

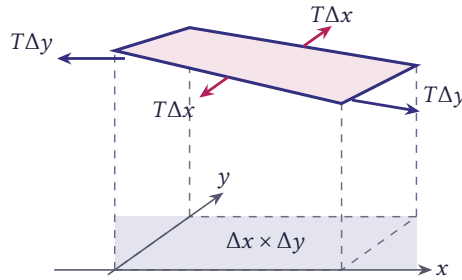


Figure 13.1 An element $\Delta x \times \Delta y$ of the vibrating membrane and the tensile forces $T\Delta x$ and $T\Delta y$ acting on it

By a modelling process analogous to the one-dimensional case, the equation of the vibrating two-dimensional membrane is obtained:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = c^2 \nabla^2 u$$

For a unique solution we need boundary conditions and initial conditions:

- Boundary conditions: $u(x, y, t) = 0$ on the boundary of the membrane
- Initial conditions: $u(x, y, 0) = f(x, y)$ and $u_t(x, y, 0) = g(x, y)$, where $u_t \triangleq \frac{\partial u}{\partial t}$

13.2 The Two-Dimensional Wave Equation with a Rectangular Boundary

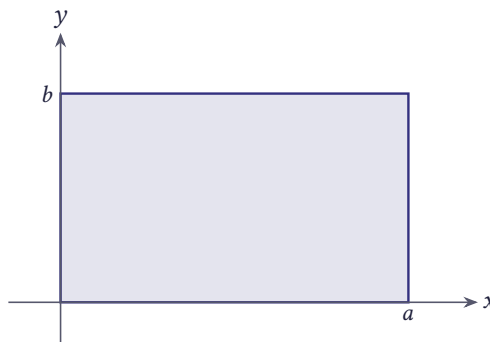


Figure 13.2 The rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

- Boundary conditions: $u(0, y, t) = u(a, y, t) = 0$ and $u(x, 0, t) = u(x, b, t) = 0$
- Initial conditions: $u(x, y, 0) = f(x, y)$ and $u_t(x, y, 0) = g(x, y)$

■ Solving the Equation

Here we solve the equation by **separation of variables**. Assume $u(x, y, t) = X(x) \times Y(y) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''YT, \quad \frac{\partial^2 u}{\partial y^2} = XY''T, \quad \frac{\partial^2 u}{\partial t^2} = XYT'' \implies XYT'' = c^2(X''YT + XY''T)$$

$$\frac{T''}{c^2T} = \frac{X''}{X} + \frac{Y''}{Y} = -\lambda^2 \implies \frac{Y''}{Y} = -\lambda^2 - \frac{X''}{X} = -\rho^2 \implies \frac{X''}{X} = -\lambda^2 + \rho^2 = -\mu^2$$

$$\begin{cases} X'' + \mu^2 X = 0 \\ Y'' + \rho^2 Y = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases} \quad \lambda^2 = \mu^2 + \rho^2$$

The solution in the x direction:

$$X(x) = A_1 \cos \mu x + A_2 \sin \mu x, \quad X(0) = 0 \implies A_1 = 0, \quad X(a) = 0 \implies A_2 \sin \mu a = 0 \implies \mu a = n\pi \implies \mu_n = \frac{n\pi}{a}$$

$$X_n(x) = A_n \sin \mu_n x, \quad \mu_n = \frac{n\pi}{a}$$

The solution in the y direction:

$$Y(y) = B_1 \cos \rho y + B_2 \sin \rho y, \quad Y(0) = 0 \implies B_1 = 0, \quad Y(b) = 0 \implies B_2 \sin \rho b = 0 \implies \rho b = m\pi \implies \rho_m = \frac{m\pi}{b}$$

$$Y_m(y) = B_m \sin \rho_m y, \quad \rho_m = \frac{m\pi}{b}$$

The solution in time:

$$\lambda_{nm}^2 = \mu_n^2 + \rho_m^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2, \quad T_{nm}(t) = C_1 \cos(c\lambda_{nm}t) + C_2 \sin(c\lambda_{nm}t)$$

$$u_{nm}(x, y, t) = [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right), \quad n, m = 1, 2, \dots$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right)$$

Applying the initial condition $u(x, y, 0) = f(x, y)$:

$$f(x, y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) = \sum_{n=1}^{\infty} K_n(y) \sin \frac{n\pi}{a}x, \quad K_n(y) \triangleq \sum_{m=1}^{\infty} A_{nm} \sin \frac{m\pi}{b}y$$

$$K_n(y) = \frac{2}{a} \int_0^a f(x, y) \sin \frac{n\pi}{a}x \, dx, \quad A_{nm} = \frac{2}{b} \int_0^b K_n(y) \sin \frac{m\pi}{b}y \, dy$$

$$A_{nm} = \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a}x \, dx \right\} \sin \frac{m\pi}{b}y \, dy$$

Exercise 13.1

Prove that

$$B_{nm} = \frac{4}{abc\lambda_{nm}} \int_0^b \left\{ \int_0^a g(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy$$

Key result 13.2 ■ Solution of the two-dimensional wave equation on a rectangle

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right), \quad \lambda_{nm}^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2$$

$$A_{nm} = \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy, \quad B_{nm} = \frac{4}{abc\lambda_{nm}} \int_0^b \left\{ \int_0^a g(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy$$

Example 13.3

Find the solution of the two-dimensional motion equation above for $g(x, y) = 0$ and $f(x, y) = xy$.

$$g(x, y) = 0 \Rightarrow B_{nm} = 0$$

$$\begin{aligned} A_{nm} &= \frac{4}{ab} \int_0^b \left\{ \int_0^a xy \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy = \frac{4}{ab} \left\{ \left[\int_0^b y \sin \frac{m\pi}{b} y dy \right] \times \left[\int_0^a x \sin \frac{n\pi}{a} x dx \right] \right\} \\ &= \frac{4}{ab} \left[y \left(\frac{-b}{m\pi} \cos \frac{m\pi}{b} y \right) - (1) \left(\frac{-b^2}{m^2\pi^2} \sin \frac{m\pi}{b} y \right) \right]_0^b \times \left[x \left(\frac{-a}{n\pi} \cos \frac{n\pi}{a} x \right) - (1) \left(\frac{-a^2}{n^2\pi^2} \sin \frac{n\pi}{a} x \right) \right]_0^a \\ &= \frac{4}{ab} \cdot \frac{-b^2}{m\pi} (-1)^m \cdot \frac{-a^2}{n\pi} (-1)^n = \frac{4ab}{nm\pi^2} (-1)^{n+m} \end{aligned}$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{4ab}{nm\pi^2} (-1)^{n+m} \cos(c\lambda_{nm}t) \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right)$$

13.3 The Two-Dimensional Heat Equation with a Rectangular Boundary

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = c^2 \nabla^2 u$$

- Boundary condition: $u(x, y, t) = 0$ on the boundary of the plate
- Initial condition: $u(x, y, 0) = f(x, y)$

■ Solving the Equation

Assume $u(x, y, t) = X(x) \times Y(y) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''YT, \quad \frac{\partial^2 u}{\partial y^2} = XY''T, \quad \frac{\partial u}{\partial t} = XYT' \implies XYT' = c^2(X''YT + XY''T)$$

$$\frac{T'}{c^2T} = \frac{X''}{X} + \frac{Y''}{Y} = -\lambda^2, \quad \frac{Y''}{Y} = -\lambda^2 - \frac{X''}{X} = -\rho^2, \quad \frac{X''}{X} = -\lambda^2 + \rho^2 = -\mu^2$$

$$\begin{cases} X'' + \mu^2 X = 0 \\ Y'' + \rho^2 Y = 0 \\ T' + c^2 \lambda^2 T = 0 \end{cases} \quad \lambda^2 = \mu^2 + \rho^2$$

As in the wave case ($X_n(x) = A_n \sin \mu_n x$ with $\mu_n = n\pi/a$ and $Y_m(y) = B_m \sin \rho_m y$ with $\rho_m = m\pi/b$), and with

$$T_{nm}(t) = C_1 e^{-c^2 \lambda_{nm}^2 t}, \quad \lambda_{nm}^2 = \mu_n^2 + \rho_m^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2$$

$$u_{nm}(x, y, t) = A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) e^{-c^2 \lambda_{nm}^2 t}, \quad n = 1, 2, \dots, m = 1, 2, \dots$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) e^{-c^2 \lambda_{nm}^2 t}$$

Applying the initial condition: exactly as in the wave case,

$$A_{nm} = \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy$$

13.4 The Laplacian in Polar Coordinates

The aim is to write $\nabla^2 u = u_{xx} + u_{yy}$ in polar coordinates in terms of r, θ .

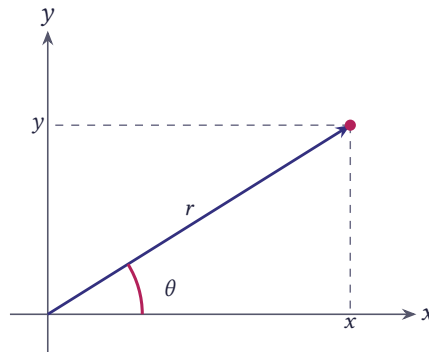


Figure 13.3 Polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$

$$r = \sqrt{x^2 + y^2}, \quad x = r \cos \theta, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right), \quad y = r \sin \theta$$

$$u_x = u_r r_x + u_\theta \theta_x \implies u_{xx} = (u_r r_x + u_\theta \theta_x)_x = u_{rr} r_x^2 + u_{\theta\theta} \theta_x^2 + 2r_x \theta_x u_{r\theta} + r_{xx} u_r + \theta_{xx} u_\theta$$

$$u_y = u_r r_y + u_\theta \theta_y \implies u_{yy} = u_{rr} r_y^2 + u_{\theta\theta} \theta_y^2 + 2r_y \theta_y u_{r\theta} + r_{yy} u_r + \theta_{yy} u_\theta$$

$$r_x = \frac{x}{r}, \quad r_{xx} = \frac{y^2}{r^3}, \quad \theta_x = \frac{-y}{r^2}, \quad \theta_{xx} = \frac{2xy}{r^4}, \quad r_y = \frac{y}{r}, \quad r_{yy} = \frac{x^2}{r^3}, \quad \theta_y = \frac{x}{r^2}, \quad \theta_{yy} = \frac{-2xy}{r^4}$$

$$u_{xx} = \frac{x^2}{r^2} u_{rr} + \frac{y^2}{r^4} u_{\theta\theta} + \frac{-2xy}{r^3} u_{r\theta} + \frac{y^2}{r^3} u_r + \frac{2xy}{r^4} u_\theta$$

$$u_{yy} = \frac{y^2}{r^2} u_{rr} + \frac{x^2}{r^4} u_{\theta\theta} + \frac{2xy}{r^3} u_{r\theta} + \frac{x^2}{r^3} u_r + \frac{-2xy}{r^4} u_\theta$$

Adding these two relations:

Key result 13.4 ■ The Laplacian in polar coordinates

$$\nabla^2 u = u_{rr} + \frac{1}{r^2} u_{\theta\theta} + \frac{1}{r} u_r$$

13.5 The Two-Dimensional Wave Equation with a Circular Boundary

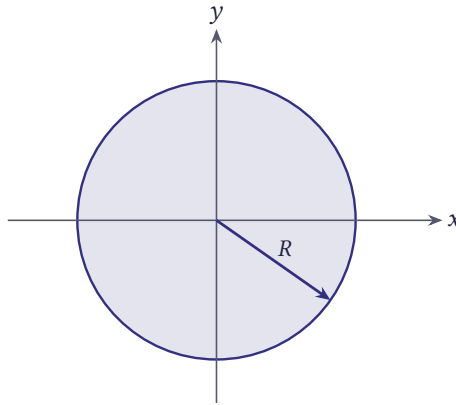


Figure 13.4 A circular membrane of radius R

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= c^2 \nabla^2 u, & u(R, t) &= 0 \text{ (boundary condition),} & u(r, 0) &= f(r), & u_t(r, 0) &= g(r) \\ & & & & & & & \text{(initial conditions)} \end{aligned}$$

■ Solving the Equation

Because of the circular symmetry of this problem it is better to solve it in polar coordinates. On a circular membrane the vibration is not a function of θ , so

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \left(u_{rr} + \frac{1}{r} u_r \right) \\ u(R, t) = 0, \quad u(r, 0) = f(r), \quad u_t(r, 0) = g(r) \end{cases}$$

Now, assuming separable variables $u(r, t) = P(r) \times T(t)$:

$$PT'' = c^2 \left(P''T + \frac{1}{r} P'T \right) \Rightarrow \frac{T''}{c^2 T} = \frac{P''}{P} + \frac{1}{r} \frac{P'}{P} = -\lambda^2 \Rightarrow \begin{cases} P'' + \frac{1}{r} P' + \lambda^2 P = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases}$$

■ Bessel's Differential Equation of Order n

$$y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0 \implies y(x) = AJ_n(x) + BY_n(x)$$

where J_n are the Bessel functions of the first kind of order n and Y_n the Bessel functions of the second kind of order n .

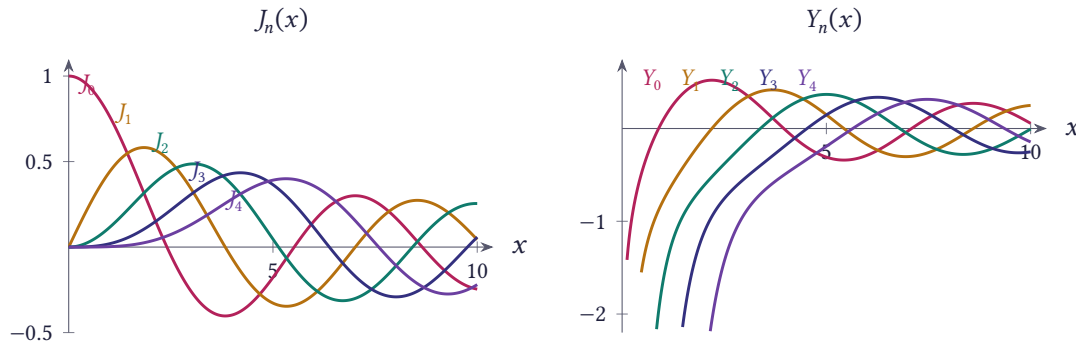


Figure 13.5 The Bessel functions of the first kind $J_0, \dots, J_4$ and of the second kind $Y_0, \dots, Y_4$ for $0 < x \leq 10$

With the substitution $s = \lambda r$:

$$P' = \frac{dP}{dr} = \frac{dP}{ds} \frac{ds}{dr} = \lambda \frac{dP}{ds}, \quad P'' = \lambda^2 \frac{d^2P}{ds^2}, \quad \frac{1}{r} \lambda = \frac{\lambda^2}{s}$$

$$\lambda^2 \frac{d^2P}{ds^2} + \frac{1}{r} \lambda \frac{dP}{ds} + \lambda^2 P = 0 \implies \frac{d^2P}{ds^2} + \frac{1}{s} \frac{dP}{ds} + P = 0$$

This is Bessel's equation of order zero ($n = 0$), so

$$P(r) = AJ_0(\lambda r) + BY_0(\lambda r)$$

Applying the boundary conditions:

- Since $P(0) \neq \infty$ and Y_0 is unbounded at the origin, the coefficient of Y_0 is zero ($D = 0$) and $P(r) = CJ_0(\lambda r)$.
- $P(R) = 0$ gives $J_0(\lambda R) = 0$, i.e. $\lambda R = \alpha_m$, where α_m are the roots of J_0 :

$$\lambda = \lambda_m = \frac{\alpha_m}{R}, \quad m = 1, 2, \dots \implies P_m(r) = C_m J_0\left(\frac{\alpha_m}{R} r\right)$$

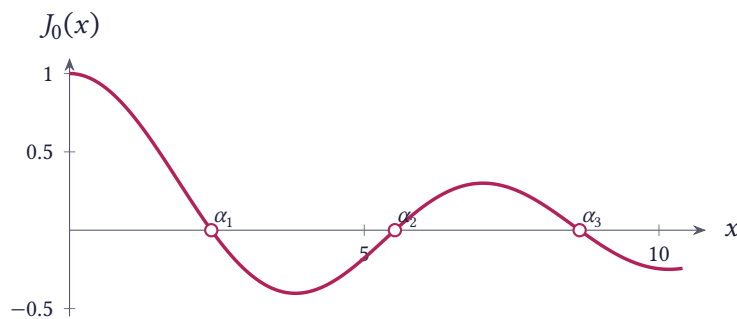


Figure 13.6 The function $J_0(x)$ and its roots $\alpha_1, \alpha_2, \alpha_3$

$$T'' + c^2 \lambda_m^2 T = 0 \implies T_m(t) = A_m \cos(c \lambda_m t) + B_m \sin(c \lambda_m t)$$

Key result 13.5 ■ Solution of the two-dimensional wave equation on a disc

$$u(r, t) = \sum_{m=1}^{\infty} (A_m \cos(c\lambda_m t) + B_m \sin(c\lambda_m t)) J_0\left(\frac{\alpha_m}{R} r\right), \quad \lambda_m = \frac{\alpha_m}{R}$$

Research task (up to 1 bonus point)

Prepare a complete report on Bessel's differential equation of order n , the Bessel functions of the first and second kind and their properties, as well as the complete solution of the two-dimensional wave equation with a circular boundary (the report must be typed, in Word format).